ESG Sentiment Classification with a Linear SVM

This model uses a traditional machine learning approach to classify ESG-related text into positive or negative sentiment. It employs a Support Vector Machine (SVM), specifically a Linear Support Vector Classification (LinearSVC) model, which is known to be highly effective and efficient for text classification tasks. The workflow has three stages: data preprocessing, feature extraction, and model training.

The `ESG_Sentiment.csv` dataset was used. Using the Pandas library, the categorical 'Sentiment' labels ("Positive" and "Negative") were mapped to binary numerical values (1 and 0, respectively) to prepare them for the model.

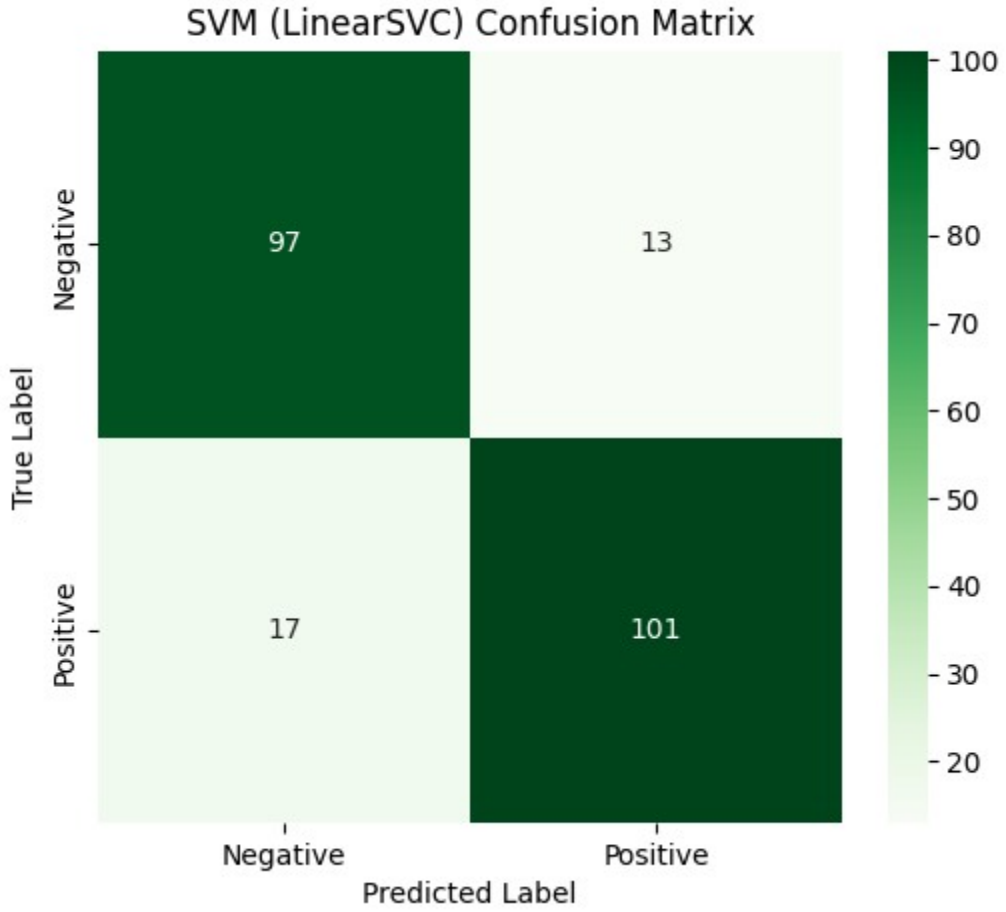
For text representation, a **TF–IDF vectorizer** was employed. It was configured to remove English stop words and limit the vocabulary to the **5,000** most informative features. A LinearSVC classifier was then trained on **75%** of the dataset, with the remaining **25%** reserved for testing. Model performance was evaluated using standard metrics, including precision, recall, and F1-score, as detailed in the classification report. Additionally, the model's coefficients were analyzed to identify the key terms that most strongly influence positive and negative sentiment.

Data and Findings

The data source is ESG_Sentiment.csv: 910 short sentences about companies' products and ESG (Environmental, Social, and Governance) practices, each labelled Positive (471) or Negative (439). The model was trained on 682 sentences and tested on the remaining 228.

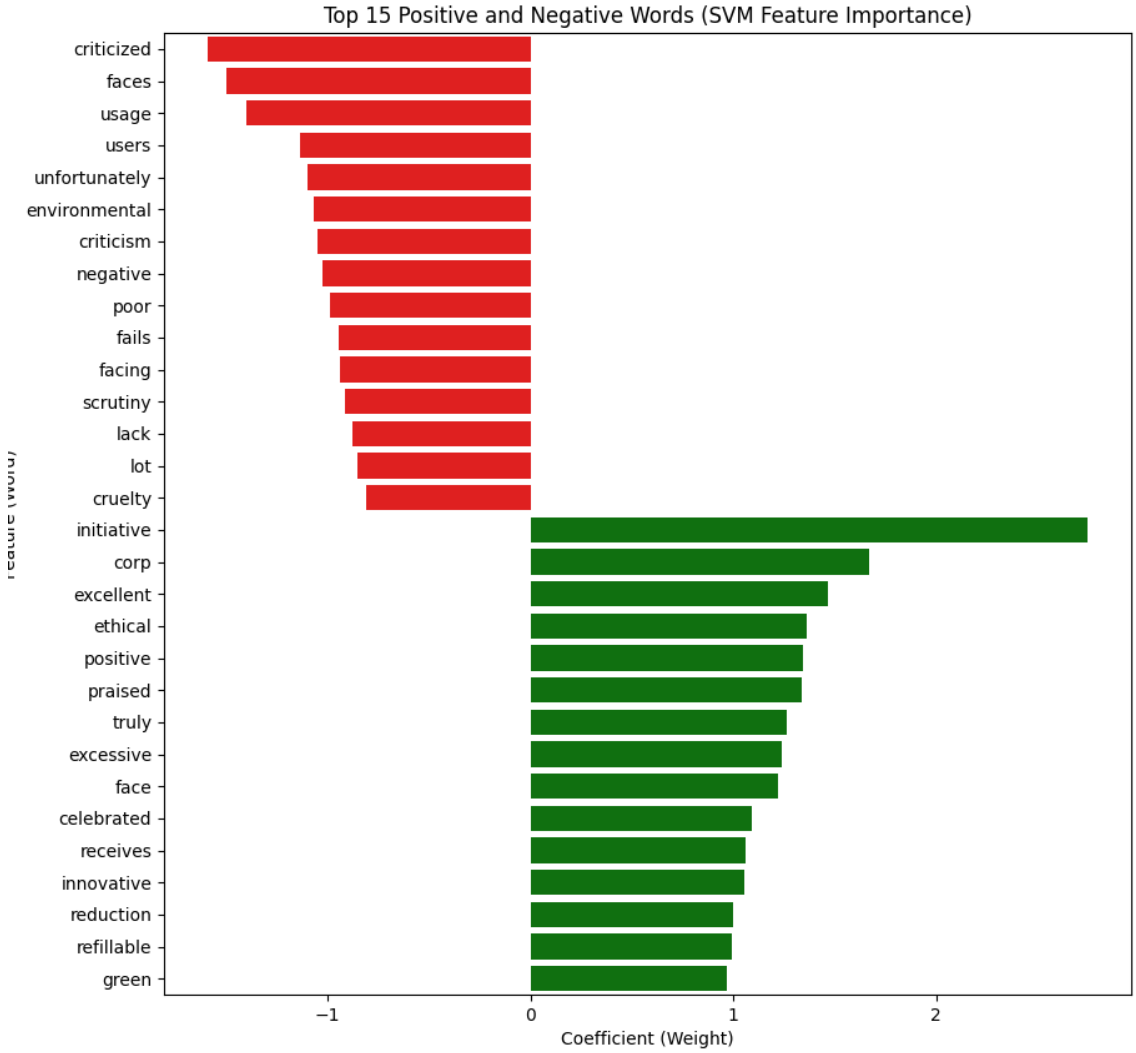
The findings from the model's evaluation are as follows:

1. **Confusion Matrix:**



On the 228 held-out test sentences the model scored 86.84% accuracy, with an F1-score of 0.87 for both classes. It misclassified 30 of 228 sentences: 13 negative statements read as positive and 17 positive statements read as negative.

2. **Feature Importance:**



The feature importance plot shows which words push a sentence towards each label. The strongest negative signals are "criticized", "faces", "criticism", "poor", "fails", "scrutiny", "lack" and "cruelty". The strongest positive signals are "initiative", "excellent", "ethical", "praised", "celebrated", "innovative", "reduction", "refillable" and "green". Some heavily weighted words, such as "corp", "usage", "users" and "excessive", carry no sentiment on their own: the model has partly learned the sentence templates of this dataset rather than the meaning. Tested on five real-world ESG sentences outside the dataset, it labelled only two correctly, which confirms that more varied training data is the next step.